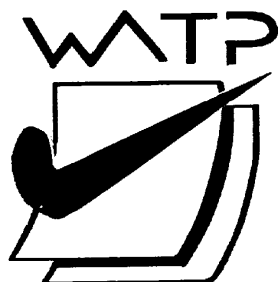


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SEMESTER TWO

**MATHEMATICS
METHODS
UNITS 1 and 2**

2017

SOLUTIONS

Calculator-free Solutions

1. Origin is at (2, 8)
 Radius = $\sqrt{(2-6)^2 + (8-8)^2}$
 = 4
 $(x-2)^2 + (y-8)^2 = 16$ ✓ ✓ [2]

2. (a) $\sin 2x = \frac{\sqrt{3}}{2}$ ✓
 $\therefore 2x = \frac{\pi}{3}, \frac{2\pi}{3}, -\frac{4\pi}{3}, -\frac{5\pi}{3}$ ✓
 $\therefore x = \frac{\pi}{6}, \frac{\pi}{3}, -\frac{2\pi}{3}, -\frac{5\pi}{6}$ ✓
 (b) $\cos x$ cannot be more than 1. ✓ [4]

3. (a) $\frac{dy}{dx} = \frac{9}{5}kx^2$ ✓✓
 (b) $y = 3 - \frac{3x^2}{2} + \frac{x^{-\frac{1}{2}}}{2}$ ✓
 $\frac{dy}{dx} = -3x - \frac{1}{4x^{\frac{3}{2}}}$ ✓✓
 (c) $y = 4x^3 - 9x$ ✓
 $\frac{dy}{dx} = 12x^2 - 9$ ✓ [7]

4.

$y = -2^{-x} - 1$	$y = \left(\frac{1}{2}\right)^x$	$y = 2^x + 3$	$y = 2^x - 1$
D	A	B	C

✓✓✓✓ [4]

5. (a) (i) $(2^4 \times 3^4)^{\frac{1}{4}}$
 = 6 ✓
 (ii) $\frac{(ab^2c^{-3})^2 \sqrt{a^4b^{-2}c^6}}{a^3b^2c}$
 = $\frac{ab}{c^{\frac{1}{2}}}$ ✓✓
 (iii) $\frac{24^{\frac{1}{3}} \sqrt[3]{18} \times 3^{\frac{1}{3}}}{\sqrt[6]{8} \cdot (27)^{\frac{1}{3}}}$
 = $\frac{(2^3 \cdot 3)^{\frac{1}{3}} \cdot 3 \cdot 2^{\frac{1}{2}} \times 3^{\frac{1}{3}}}{(2^3)^{\frac{1}{6}} \cdot 3}$ ✓✓
 = $2^2 \times 3$
 = 12 ✓

(b) (i) $\frac{2^{4x-4}}{2^{-4}} = 2^{-3}$ ✓
 $2^{4x} = 2^{-3}$ ✓
 $4x = -3$
 $x = -\frac{3}{4}$ ✓
(ii) $(2^x - 8)(2^x - 1) = 0$
 $2^x = 8$ or $2^x = 1$ ✓
 $x = 3$ or $x = 0$ ✓ [11]

6. (a) (i) 0.2 ✓
(ii) 0.4 ✓
(iii) 0.7 ✓
(b) ${}^5C_2 \times 2^3 \times 3^2 = 720$ ✓✓ [5]

7. (a) (i) $\int x^2 + 1 dx = \frac{x^3}{3} + x + c$ ✓✓
(ii) $\int t^{-2} - 2t + \pi t^{-3} dt$ ✓
 $= -\frac{1}{t} - t^2 - \frac{\pi}{2t^2} + c$ ✓✓
(b) $y = 3x + \frac{x^2}{2} - \frac{2x^5}{5} + c$ ✓
 $2 = 3 + \frac{1}{2} - \frac{2}{5} + c$
 $\therefore c = -\frac{11}{10}$ ✓
 $y = -\frac{2x^5}{5} + \frac{x^2}{2} + 3x - \frac{11}{10}$ ✓ [8]

8. (a) $\frac{64}{2}(4 - b) = 0$
 $b = 4$ ✓
(b) $p(x) = \frac{x^4}{2} - 2x^3$
 $p'(x) = 2x^3 - 6x^2 = 0$ ✓
 $2x^2(x - 3) = 0$
 $x = 0$ or $x = 3$ ($x = 0$ is not a minimum) ✓
When $x = 3$ $y = \frac{3^4}{2} - 2(3^3)$
 $\left(3, -\frac{27}{2}\right)$ ✓
(c) Stationary point at $x = 0$, $p'(0) = 0$
 $p''(x) = 6x^2 - 12x = 0$ ✓
 $6x(x - 2) = 0$
 $x = 0, 2$
 $\therefore x = 0$ $p'(x) = p''(x) = 0$ ✓ [6]
Horizontal point of inflection
Or sign table:

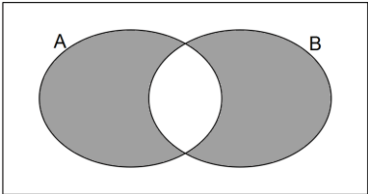
	$x < 0$	$x = 0$	$0 < x < 3$	$x = 3$	$x > 3$
$p(x)$	↓	↔	↓	↔	↑
$p'(x)$	-	0	-	0	+

9. (a) $a_1 = 3$
 $a_2 = 6$
 $a_3 = 10$ ✓✓
 (b) $T_{n+2} = T_n + T_{n+1}$ $T_1 = 1$ $T_2 = 1$ ✓ [3]

Calculator-assumed Solutions

10. (a) $x^2 = 4x^2 + 9 - 2(2x)(3)\cos Q$
 $\frac{-3(x^2 + 3)}{-12x} = \cos Q$ ✓
 $\cos Q = \frac{x^2 + 3}{4x}$ ✓
 (b) $\cos Q = 0.9125$ ✓
 $Q = 24.1468^\circ$ ✓
 $\text{Area} = \frac{1}{2}(3)(4.8)\sin 24.1468$
 $= 2.945 \text{ units}^2$ ✓ [5]

11. (a) $-\frac{b}{2a} = -\frac{2}{2a} = -\frac{1}{a}$
 $y\left(-\frac{1}{a}\right) = a - \frac{1}{a} \text{ or } \frac{a^2 - 1}{a}$
 Turning point $\left(-\frac{1}{a}, \frac{a^2 - 1}{a}\right)$ ✓✓
 (b) $b^2 - 4ac > 0$
 $4 - 4a^2 > 0$ ✓
 $a^2 < 1$
 $-1 < a < 1$ ✓ [4]

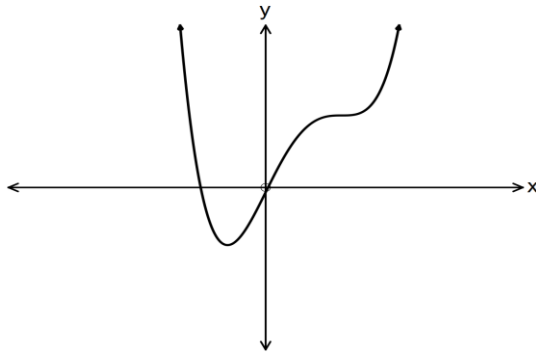
12. (a) 

- (b) (i) 0.1 ✓
 (ii) 0.2 ✓
 (iii) 0.3 ✓
 (iv) 0.9 ✓ [5]

13. (a) $C(t) = \frac{t^3}{3} - \frac{t^2}{2} - 12t + 105$ ✓
 $C(2) = 81.7 \text{ cents}$ ✓
 (b) $(t - 4)(t + 3) = 0$
 $t = 4 \text{ weeks}$ ✓
 $C(4) = 70.3 \text{ cents}$ ✓
 (c) Week 6.8 therefore during week 7 ✓ [5]

14. (a) $d = k\sqrt{h}$
 $4665 = k\sqrt{1.7}$
 $\therefore k = 3577.89 \approx 3578$ ✓
 $d = 3577.89\sqrt{(1.76 + 85)}$
 $d = 33\,326.3 \text{ m}$ ✓
 $d \approx 33 \text{ km}$ ✓
- (b) $d = 3577.89\sqrt{0.8h}$ ✓
 $d = 3577.89(0.8944)\sqrt{h}$ ✓
 d decreases by 10.6%. ✓ [5]
15. (a) (i) $T_{n+1} = T_n + \sqrt{2}$ $T_1 = 3 - \sqrt{2}$ ✓✓
(ii) $T_{n+1} = 3 T_n$ $T_1 = -2$ ✓✓
- (b) $T_n = 17 + (n - 1)(3)$ ✓
 $T_{25} = 17 + (24)(3)$
 $= 89$ ✓ [6]
16. (a)
- | $h \rightarrow 0$ | Limit = |
|-------------------|-----------|
| 0.1 | 3.05 |
| 0.01 | 3.005 |
| 0.0001 | 3.00005 |
| 0.000001 | 3.0000005 |
- $\therefore \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} = 3$ ✓✓
✓
- (b) $f(x) = 3x^2$ ✓
 $\therefore f'(x) = 6x$ ✓
- (c) $4x + k = 0$ ✓
 $4(3) + k = 0$ ✓
 $\therefore k = -12$ ✓
- (d) $y = \int 4x + 1 \, dx$
 $\therefore y = 2x^2 + x + c$ ✓
 $2(1)^2 + 1 + c = -2$
 $c = -5$
 $y = 2x^2 + x - 5$ ✓ [9]
17. (a) $\frac{\sqrt{1.44 \times 10^6}}{(2 \times 10^{-2})^4}$
 $= \frac{1.2 \times 10^3}{1.6 \times 10^{-7}}$ ✓
 $= 0.75 \times 10^{10}$ ✓
 $= 7.5 \times 10^9$ ✓
- (b) (i) $1.5 \times 10^4 = 1200(r)^6$ ✓
 $P_0 = 1200$ ✓
 $r = 1.5234$ ✓
- (ii) $P = 187500$ ✓
(iii) $t = 15.987$ ✓
15 hours and 59 minutes. ✓ [8]

18. (a)



✓✓✓

(b)

A B C
 (iii) (ii) (i)

✓✓✓ [6]

19. (a) (i) $n = 13$
 $T_{13} = 26 + (12)(2)$
 $= 50$ seats
 (ii) $n = 23$ Row W
 (iii) $n = 24$
 Sum of seats = 1176

✓

✓

✓

✓

- (b) $205 = 5(2a + 9d)$
 $710 = 10(2a + 19d)$
 $a = 7 \quad d = 3$
 $S_{30} = 15(14 + 29(3))$
 $= 1515$

✓

✓

✓

- (c) $a = 254 \quad d = -3 \quad T_n = 176$
 $176 = 254 + (n - 1)(-3)$
 $n = 27$
 $S_n = \frac{27}{2}(254 + 176)$
 $= 5805$

✓

✓

✓ [10]

20. (a) $V(x) = 3x(90 - 3x)\left(\frac{x}{3}\right)$
 $= x^2(90 - 3x)$
 $= 90x^2 - 3x^3$

✓

✓

- (b) $V'(x) = 180x - 9x^2$
 $9x(20 - x) = 0$
 $x = 0$ or $x = 20$
 $V''(x) = 180 - 18x \quad V''(20) < 0 \therefore$ Maximum
 Maximum volume is 12000 m^3
 When length = 60 m , width = 30 m and depth = 6.67 m

✓

✓

✓

✓ [6]

21. (a) (i) $A_1 = \frac{3}{2}$ $A_2 = \frac{9}{2}$ $A_3 = \frac{27}{2}$ ✓
- (ii) $A_{n+1} = 3 A_n$ $A_1 = \frac{3}{2}$ ✓✓
- (iii) $S_{25} = 6.355 \times 10^{11}$ ✓✓
- (b) (i) $T_n = 2 T_{n-1}$ $T_0 = 6$
 $T_{12} = 24\,576$ users ✓✓
- (ii) After 18 months
 (Or during the 18th month) ✓✓
- (iii) $\frac{6000000}{1.15} = 5\,217\,391$ users ✓
- During the 20th month ✓
- (c) $S_\infty = \frac{a}{1-r}$
 $= \frac{45}{100} \div \left(1 - \frac{1}{100}\right)$ ✓
 $= \frac{45}{99}$
 $= \frac{5}{11}$ ✓ [13]
22. (a) $s = -\frac{t^3}{2} + t^2 + 20t$. $s(3) = 55.5$ m ✓
- (b) $\frac{55.5}{3} = 18.5$ m/s ✓
- (c) $v = -\frac{3t^2}{2} + 2t + 20$ ✓
 $v(2) = 18$ m/s ✓
- (d) $-\frac{3t^2}{2} + 2t + 20 = 0$ ✓
 $t = 4.38$ s ✓
 Max displacement: $s(4.38) = 64.8$ m ✓
- (e) $s(8) = -32$ ✓
 Total distance = $64.8 + 64.8 + 32 = 161.6$ m ✓ [9]
23. (a) $g'(x) = 2x - 6$
 $m = \frac{1}{3} \therefore 2x - 6 = \frac{1}{3}$ ✓
 $x = \frac{19}{6} = 3\frac{1}{6}$ ✓
 $y = \frac{1}{36} \left(\frac{19}{6}, \frac{1}{36}\right)$ ✓
- (b) (i) $f'(x) = x^2 + 2ax + b$
 $36 - 12a + b = 0$ ✓
 $-72 + 36a - 6b = 0$ ✓
 Solving simultaneously $a = 4$ $b = 12$ ✓
- (ii) $x^2 + 8x + 12 = 0$
 $x = -6$ or $x = -2$ ✓
 Turning point $\left(-2, -\frac{32}{3}\right)$ ✓
 $f''(x) = 2x + 8$ $f''(-2) > 0$ Therefore Local Minimum ✓ [9]

End of Questions